Physics Simulation and STRidge Identification Report (2D Lid Cavity -- U equation)

> Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	1.00000000e+03	kg m^{-3}
Dynamic viscosity	μ	1.00000000e+01	Pa s
Kinematic viscosity	ν	1.00000000e-02	m^2 s^{-1}
Characteristic velocity	U	1.00000000e+00	m s^{-1}
Characteristic length	D	1.00000000e+00	m
Reynolds number	$Re = \rho U D/\mu$	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s^{-2}

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
Grid resolution	$N_x \times N_y$	31 x 31	cells
Grid spacing	$\Delta x, \Delta y$	3.22580645e-02, 3.22580645e-02	m
Time step	Δt	1.00000000e-04	s
Numerical threshold	ϵ	7.00000000e-11	-
Advective coefficient	c_a	1.00000000e+00	-
Pressure coefficient	$c_p = 1/\rho$	1.00000000e-03	m^3 kg^{-1}
Viscous coefficient	$c_v = 1/Re$	1.00000000e-02	-

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(6655, 30, 31)
Regression field shape (t, x, y)	(4255, 22, 27)
Regression samples	2,527,470
Dictionary terms	5

Quantity	Value
First saved time index	150
Temporal crop at each boundary	1200 layers
x-boundary crop	4 points
y-boundary crop	2 points

4. Governing Equation Verification

Reference equation

`u_t =- 1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_xx + 1.00000000e-02*u_yy - 1.00000000e-03*p_x`

Learned equation

`u_t =- 1.13842289e+00*uu_x - 1.03055812e+00*vu_y + 8.40649079e-03*u_xx + 1.03461437e-02*u_yy - 1.04792514e-03*p_x`

Term	Reference coefficient	Learned coefficient	Absolute error
uu_x	-1.00000000e+00	-1.13842289e+00	1.38422889e-01
vu_y	-1.00000000e+00	-1.03055812e+00	3.05581219e-02
u_xx	+1.00000000e-02	+8.40649079e-03	1.59350921e-03
u_yy	+1.00000000e-02	+1.03461437e-02	3.46143665e-04
p_x	-1.00000000e-03	-1.04792514e-03	4.79251383e-05

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	9.42668455e-04
L0 penalty	1.00000000e-06
Training RMSE	1.10494359e-02
Validation RMSE	6.16076275e-03

6. Reproducibility Notes

- Input directories: `/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_lidcav`
- Velocity and pressure fields were loaded starting at saved time index `150`.

- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the reference Navier-Stokes coefficients defined in this script.